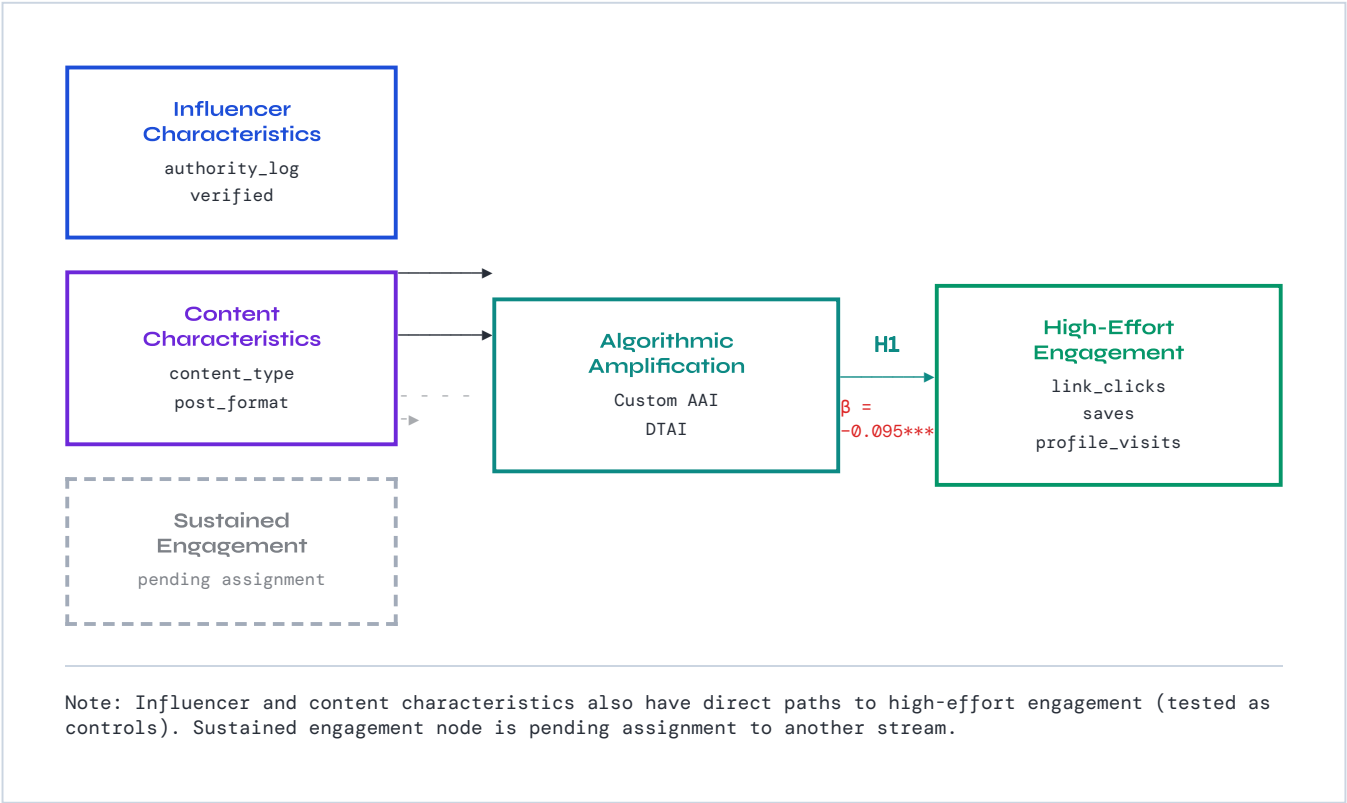


Does the Algorithm Drive High-Effort Engagement?

Full results for the algorithmic amplification research stream — testing whether algorithmic amplification features are associated with high-effort engagement (clicks, saves, profile visits) across 10,000 social media posts from 800 influencers.

01 Research Question & Model Structure

RQ1: To what extent are algorithmic amplification features associated with high-effort engagement?



What this shows: The model tests whether algorithmic amplification — measured by a custom composite index and a binary indicator — is associated with high-effort engagement after accounting for who the influencer is and what type of content they post. The input nodes feed the algorithm, and the algorithm feeds the outcome.

02 How the Custom AAI Was Built

Rather than using the provided black-box amplification index, a custom AAI was built from four theoretically grounded components — each z-scored and averaged equally. This allows every component to be named, explained, and defended.

$$\text{FORMULA} \quad \text{Custom AAI} = z(\text{Breadth}) + z(\text{Depth}) + z(\text{Momentum}) + z(\text{Persistence}) \div 4$$

01 · BREADTH

Breadth

reach_to_follower

reach /
follower_count

How far a post traveled beyond the creator's own followers. A ratio above 1.0 means the algorithm pushed it to new audiences.

r = -0.32

02 · MOMENTUM

Momentum

reach_growth_rate

proxy 0-1

How actively the post was being re-engaged over time. Values closer to 1 mean sustained audience activity beyond the initial burst.

r = +0.29

03 · DEPTH

Depth

impressions_per_reach

impressions /
reach

How many times each unique viewer was shown the post on average. Repeated exposure does not meaningfully drive high-effort action.

r = +0.00

04 · PERSISTENCE

Persistence

exposure_persistence_hours

hours in circulation

How long the post stayed active in platform feeds. Longer circulation may reflect repeated surfacing to audiences who already chose not to act.

r = -0.11

What this means: Two components pull negative (breadth and persistence), one pulls positive (momentum), and one has no effect (depth). When combined, breadth dominates and the composite ends up negatively associated with high-effort engagement overall. The only amplification mechanism that helps is momentum — posts gaining reach quickly from motivated viewers.

03 The Dual-Threshold Amplification Indicator (DTAI)

A binary indicator was created to identify posts where the algorithm did two things simultaneously — pushed content beyond the creator's organic follower base AND repeatedly surfaced it to that expanded audience. These posts represent the most intensive algorithmic amplification in the dataset.

DTAI DEFINITION

DTAI = 1 if Both Conditions Are Met**Condition 1 — Breadth**

reach / follower_count ≥ median (0.36)
Post reached more people relative to follower base than the typical post

Condition 2 — Depth

impressions_per_reach ≥ median (1.61)

WHY THESE TWO CONDITIONS

Both conditions are **ratio-based and time-independent** — neither depends on a measurement window, so there is no time period to defend.

Both come directly from the **AAI's breadth and depth components**, making DTAI theoretically consistent with the continuous index.

Each viewer was shown the post more times than average

Using the **median as the threshold** means the cutpoint has a clear interpretation: above average on both amplification dimensions simultaneously.

POSTS CLASSIFIED DTAI = 1

24.9%

2,491 out of 10,000 posts met both conditions — above median on breadth AND depth of amplification

HE: DTAI = 1 vs DTAI = 0

-51.4%

DTAI = 1 posts averaged 2.26 clicks/saves/visits vs 4.66 for DTAI = 0 posts. Heavy amplification cut meaningful engagement in half.

REGRESSION RESULT

$\beta = -0.090$

After controlling for influencer authority, verified status, content type, and post format — DTAI remains significant ($p < .001$)

What this means: When the platform both expanded a post's reach beyond its organic audience and kept showing it to viewers repeatedly, those posts generated about half the clicks, saves, and profile visits of posts that didn't cross both thresholds. The algorithm doing extra work is actively working against conversion.

04 Core Finding — AAI → High-Effort Engagement

More algorithmic amplification is associated with **fewer** clicks, saves, and profile visits — both before and after controlling for influencer and content characteristics.

NO CONTROLS

$\beta = -0.138*$**

CORRELATION $r = -0.071$ $R^2 = 0.005$ $p\text{-value} < 0.001$

WITH CONTROLS

$\beta = -0.095*$**

$R^2 = 0.462$ $CI [-0.129, -0.061]$ $p\text{-value} < 0.001$

What this means: The negative relationship between algorithmic amplification and high-effort engagement holds both raw and after accounting for everything else in the model. A one standard deviation increase in AAI is associated with about 9.5% lower high-effort engagement. The algorithm reaching more people does not translate into more motivated audience behavior.

05 The Influencer vs The Algorithm

When influencer authority and algorithmic amplification are compared directly, **the influencer wins by a landslide** — explaining 91x more variance in high-effort engagement than the algorithm does.

PREDICTOR	CORRELATION	β (CONTROLLED)	R ² ALONE	DIRECTION
authority_log influencer size	+0.676	+0.586***	45.7%	POSITIVE
custom_aai algorithmic amplification	-0.071	-0.095***	0.5%	NEGATIVE
verified platform credibility	+0.285	+0.045*	—	POSITIVE

VARIANCE EXPLAINED (R²) IN HIGH-EFFORT ENGAGEMENT

Influencer characteristics alone	45.7%
Full model (all controls + AAI)	46.2%
Custom AAI alone	0.5%
Content characteristics alone	0.2%

Adding AAI and content characteristics to the influencer model improves R² by only 0.5%. The influencer is doing almost all the work.

What this means: Knowing who the influencer is explains 45.7% of why some posts get more clicks and saves than others. Knowing how widely the algorithm distributed the post explains just 0.5%. The algorithm follows the influencer — high-authority accounts naturally receive both more amplification and more high-effort engagement, but when you separate these effects, only authority drives motivated behavior.

06 Does Influencer Size Change How Amplification Works?

Yes — and significantly. The larger the influencer, the more algorithmic amplification **hurts** high-effort engagement. The interaction between AAI and authority is the strongest moderation finding in the analysis.

Model 1: AAI × Influencer → High-Effort Engagement

clicks, saves, profile visits

R²
0.467

BASE EFFECT OF AAI (mean-centered)
 β = -0.102 — amplification hurts high-effort engagement

AAI × Authority
Does being a bigger influencer change how amplification affects HE? **-0.112** p < .001 *******

AAI × Verified
Does having a verification badge change how amplification affects HE? **-0.012** p = .780 **ns**

The bigger the influencer, the more amplification hurts their high-effort engagement. Their followers are already motivated — algorithmic reach expansion adds disengaged strangers instead. Whether the account has a verification badge makes no

significant difference to how amplification affects high-effort engagement.

What this means: A large influencer's motivated follower base is already doing the conversion work. When the algorithm expands reach beyond that core audience, it dilutes the quality of new viewers. For a small influencer the damage is less severe because there are fewer motivated followers to begin with — the baseline is already lower. Verification only matters for passive likes, not for clicks and saves.

07 Does Content Type or Format Change Anything?

No. Whether a post is promotional or informational, video or image, **makes no difference** to how algorithmic amplification affects high-effort engagement.



No Moderation Found — The Algorithm Treats All Content the Same

All eight interaction tests between AAI and content characteristics came back non-significant. Content type and post format do not change how algorithmic amplification relates to high-effort engagement.

AAI × promotional vs informational → HE ns	AAI × experiential vs informational → HE ns
AAI × video vs image → HE ns	AAI × carousel vs image → HE ns

What this means: Changing content strategy — switching to video, making posts more promotional, experimenting with carousels — will not change how algorithmic reach affects whether people click, save, or visit a profile. The algorithm responds to engagement signals, not to what the content actually is. This means the audience mismatch problem cannot be fixed through content alone.

08 Are Influencer and Content Characteristics Related to Each Other?

No. Large influencers do not systematically post different content types or formats than small influencers. The two input nodes are **completely independent** — their contributions to the model do not overlap.

Experiential Content vs informational $r(\text{authority}) = -0.010$	Promotional Content vs informational $r(\text{authority}) = +0.015$	Video Format vs image $r(\text{authority}) = +0.006$	Carousel Format vs image $r(\text{authority}) = -0.007$
--	---	--	---

$r(\text{verified}) = -0.005$
 $R^2 = 0.000$

$r(\text{verified}) = +0.006$
 $R^2 = 0.000$

$r(\text{verified}) = +0.008$
 $R^2 = 0.000$

$r(\text{verified}) = -0.020$
 $R^2 = 0.000$

What this means: The influencer findings and content findings are genuinely telling us separate things. The fact that big influencers drive more high-effort engagement is not because big influencers happen to post more video or more promotional content. The two nodes contribute independently, which strengthens the validity of the overall model.

KEY TAKEAWAYS — ISAAC'S STREAM

01 — More Reach, Less Action

Algorithmic amplification is **negatively associated** with high-effort engagement ($\beta = -0.095, p < .001$). Posts receiving the most intensive amplification (DTAI = 1) generated 51.4% fewer clicks, saves, and profile visits. The algorithm delivers exposure, not intent.

03 — Authority Makes It Worse

The larger the influencer, the more amplification **damages** high-effort engagement ($\beta = -0.112, p < .001$). Bigger audiences have more motivated followers — reach expansion beyond that core adds disengaged strangers.

02 — The Influencer Dominates

Influencer authority alone explains **45.7% of variance** in high-effort engagement versus just 0.5% for AAI. Who posts something matters far more than how widely the platform distributes it.

04 — Content Strategy Changes Nothing

Content type and post format **do not moderate** the amplification–engagement relationship. The algorithm is content-agnostic. The two input nodes are fully independent of each other.

*** $p < .001$ · ** $p < .01$ · * $p < .05$ · ns = not significant | OLS regression · DV = $\log(1 + \text{high_effort_engagement})$ | Custom AAI = average of z-scored breadth, depth, momentum, persistence | DTAI = $\text{reach/follower} \geq \text{median}(0.36)$ AND $\text{impressions/reach} \geq \text{median}(1.61)$ | SEs clustered by influencer_id · n = 10,000 posts · 800 influencers